Jierui Li

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EDUCATION

Xidian University & Virginia Tech Joint Program Sep 2023 – Expected Jun 2027

- B.S. in Big Data Management and Application
- GPA: 3.8/4.0
- Relevant Coursework: Linear Algebra, Probability Theory, Database, Python, Data Mining

RESEARCH INTERESTS

Embodied AI, Vision-Language Robot Learning, Long-Horizon Task Reasoning, Interactive Multimodal Agents

RESEARCH EXPERIENCE

BLENDER Lab, University of Illinois Urbana-Champaign Apr 2026 – Present

Advised by Prof. Heng Ji

- Proposed FurniAnchor, a zero-shot vision-language reasoning framework for manual-guided furniture assembly, converting instruction manuals into atomic assembly operations and enabling target-free 6D pose reasoning; co-first-authored manuscript under review at AAAI 2026.
- Conceptualized a multi-agent procurement framework for biomedical laboratory equipment.

Natural Language Processing Laboratory, Tsinghua University Sep 2025 – Apr 2026

Advised by Prof. Zhiyuan Liu and Prof. Xin Yi

- Proposed RMHackBench, a mechanism-oriented benchmark for diagnosing reward hacking in vision-language reward models; manuscript under review through ACL Rolling Review (August 2026 cycle).

HPC-AI Lab, National University of Singapore Dec 2025 – Feb 2026

Advised by Prof. Yang You

- Contributed to the development of GroupToM-Bench, a benchmark for evaluating group-level Theory of Mind in multi-agent and multimodal large language models; accepted to the ACL 2026 Main Conference as an Oral paper.

National Key Laboratory of Cognitive Intelligence, University of Science and Technology of China Aug 2025 – Sep 2025

Mentored by Dr. Zhi Zheng

- Proposed a fine-grained drift modeling framework based on unmixed representations for online time series forecasting under delayed feedback; manuscript under review at WSDM 2026.

Laboratory of Intelligent Perception and Image Understanding, Xidian University Dec 2024 – Oct 2025

Advised by Prof. Hao Zhu

- Proposed RouteGraph-Mona, a parameter-efficient fine-tuning framework for mineral image classification that combines sample-adaptive routing, class-wise route anchors, and confusion-aware route-space regularization; accepted at PRICAI 2026.
- Contributed to the design of SFPN (Shape Feature Preservation Network) for shape-aware feature modeling in building change detection; manuscript under review at TGRS.

PUBLICATIONS AND MANUSCRIPTS

GroupToM-Bench: Benchmarking Group Theory of Mind and Nonlinear Social Emergence in MLLMs

ACL Main Conference 2026 *Oral*

Weidong Tang, **Jierui Li**, Yueling Hou, Zihan Mei, Zhang Can, Xinyan Wan, Zhiyuan Liang, Pengfei Zhou, Wangbo Zhao, Yang You*

We propose GroupToM-Bench, the first multimodal multi-agent benchmark for group-level Theory of Mind in MLLMs, with a seven-level cognitive framework for reasoning from individual intentions to collective outcomes.

RouteGraph-Mona: Confusion-Aware Routing Fine-Tuning for Mineral Image Classification

PRICAI 2026 *Accepted*

Jierui Li†, Zhiyuan Qi†, Hao Zhu, Yufan Liu, Jixian Liu, Shaojie Jiang, Yaqi Liu, Xiaotong Li*, Wei Wang*

We propose RouteGraph-Mona, a parameter-efficient fine-tuning framework for mineral image classification. Built upon Mona, it replaces static multi-scale aggregation with sample-adaptive routing and regularizes routing signatures using class-wise route anchors and an online confusion graph, improving separation among visually confusing mineral categories across multiple public benchmarks.

FurniReason: Zero-Shot Vision-Language Reasoning for Manual-Guided Furniture Assembly

AAAI 2027 *Under Review*

Zhiyuan Qi†, **Jierui Li**†, Yifan Shen, Cheng Qian, Jiateng Liu

We propose FurniReason, a zero-shot framework for manual-guided furniture assembly that recovers atomic assembly operations from manuals and performs target-free 6D pose reasoning through multi-view candidate-state refinement.

Fine-Grained Drift Modeling via Unmixed Representations for Online Time Series Forecasting

WSDM 2026 *Under Review*

Jierui Li, Weidong Tang, Jixian Liu, Shuning Zhang, Zhi Zheng*, Lingxing Kong

We propose Unmix Drift, a fine-grained drift modeling framework for online time series forecasting that addresses concept drift under feedback delay by decomposing non-stationary series into latent concept prototypes and leveraging unmixed representations to detect drift direction and intensity before labels arrive.

CueShiftBench: How Heuristic Cues Shape Multimodal Reward Evaluation

ACL ARR 2026 August Submission *Under Review*

Jierui Li, Yueling Hou, Zhiyuan Qi, Bo Zhang, Ziqi Ma, Jixian Liu, Wenyue Ban, Shuning Zhang, Yifan Shen, Weidong Tang, Qian Jiang, Xin Yi, Lingxing Kong

We propose CueShiftBench, a benchmark for systematically analyzing how heuristic cues shape multimodal reward evaluation and revealing shortcut-driven behaviors in vision-language reward models.

FedOLoRA: Demand-Aware Recycling of Fixed-Capacity Orthogonal Memory for Federated Continual LoRA

AAAI 2027 *Under Review*

Zhuodong Liu†, **Jierui Li**†, Tianzuo Yuan, Xiangyu Li

We propose FedOLoRA, a federated class-incremental learning framework that formulates orthogonal protection as fixed-capacity memory allocation for continual LoRA. It registers gauge-invariant product-space directions, estimates incoming-client demand, and dynamically recycles low-value protected subspaces under a hard stream-independent budget, preserving model plasticity while substantially reducing protection-state storage.

CoTAC: Collaborative Contextualization, Tokenization, and Calibration for Multimodal Generative Recommendation

AAAI 2027 — *Under Review*

Zhuodong Liu†, Yunhe Ni†, **Jierui Li**†, Yuchen Zhang, Peiyu Hu

We propose CoTAC, a multimodal generative recommendation framework that addresses the content–collaboration mismatch by carrying shared collaborative evidence through item representation, Semantic-ID tokenization, and autoregressive generation. It combines neighbor-aware multimodal contextualization, compatibility-regularized residual quantization, and prefix-conditioned soft alignment, achieving consistent improvements across four Amazon benchmarks.

Shape Feature Preservation Network for Building Change Detection

IEEE Transactions on Geoscience and Remote Sensing — *Under Review*

Xiaotong Li*, Jianda Wang, Haoyu Wang, Hao Zhu, **Jierui Li**, Huihui Dong, Biao Hou

We propose a novel network named SFPN, which effectively addresses jagged edges and misclassification caused by interfering backgrounds in building change detection through multi-scale feature enhancement and a collaborative dual-task design.